MATH 345 Skeleton Notes
Unit 4: Orthogonality and least squares
Section 4.1: Orthogonality and orthogonal bases

Basic definitions

Definition Recall the definition of the dot product of two vectors, introduced in Dot product, and the length of a vector as introduced in Length of a vector. The distance between two points $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{y} = (y_1, \cdots, y_n)$ is the length of the vector $\mathbf{x} - \mathbf{y}$, i.e.

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\| = \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2}.$$

If $\mathbf{x}$ is any non-zero vector, then the vector $\mathbf{u} = \mathbf{x}/\|\mathbf{x}\|$ is the unit vector in the direction of $\mathbf{x}$ (a unit vector is a vector of length one). Two vectors $\mathbf{x}$ and $\mathbf{y}$ are orthogonal or perpendicular if $\mathbf{x} \cdot \mathbf{y} = 0$.

Theorem: The Cauchy-Schwarz inequality If $\mathbf{x}, \mathbf{y}$ are vectors in $\mathbb{R}^n$,

$$|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\|\|\mathbf{y}\|.$$

Moreover, $|\mathbf{x} \cdot \mathbf{y}| = \|\mathbf{x}\|\|\mathbf{y}\|$ only when $\mathbf{x}$ and $\mathbf{y}$ are scalar multiples of each other.

Activity: Cosine similarity revisited

Let $\mathbf{u}$ and $\mathbf{v}$ be nonzero vectors in $\mathbb{R}^n$. The cosine similarity from Unit 1 was

$$\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}.$$

1. Use the Cauchy-Schwarz inequality to explain why this number is between -1 and 1.
2. What condition makes the cosine similarity equal to 0?
3. In Unit 1, this number measured cosine similarity. What does a value of 0 say geometrically?

Corollary: The triangle inequality If $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, then

$$\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|.$$

Remark The triangle inequality for vectors implies the triangle inequality for the distance function, i.e. that

$$d(\mathbf{x}, \mathbf{z}) \leq d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z}).$$

Indeed, we calculate that

$$
\begin{aligned}
d(\mathbf{x}, \mathbf{z}) &= \|\mathbf{x} - \mathbf{z}\| \\
&= \|\mathbf{x} - \mathbf{y} + \mathbf{y} - \mathbf{z}\| \\
&\leq \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y} - \mathbf{z}\|.
\end{aligned}
$$

Theorem Let $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{z}$ be vectors in $\mathbb{R}^n$, and let a be a scalar in $\mathbb{R}$. Then the following properties hold:

- $\mathbf{x} \cdot \mathbf{y} = \mathbf{y} \cdot \mathbf{x}$.
- $\mathbf{x} \cdot (\mathbf{y} + \mathbf{z}) = \mathbf{x} \cdot \mathbf{y} + \mathbf{x} \cdot \mathbf{z}$.
- $(a\mathbf{x}) \cdot \mathbf{y} = a(\mathbf{x} \cdot \mathbf{y}) = \mathbf{x} \cdot (a\mathbf{y})$.
- $\|\mathbf{x}\|^2 = \mathbf{x} \cdot \mathbf{x}$.
- $\mathbf{x} \cdot \mathbf{x} \geq 0$, and $\mathbf{x} \cdot \mathbf{x} = 0$ if and only if $\mathbf{x} = \mathbf{0}$.
- $\|a\mathbf{x}\| = |a|\|\mathbf{x}\|$.

Activity

Show that $\|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + 2(\mathbf{x} \cdot \mathbf{y}) + \|\mathbf{y}\|^2$ for any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.

Activity

Suppose that $\mathbb{R}^n = \mathrm{span}\{\mathbf{f}_1, \dots, \mathbf{f}_k\}$ for some vectors $\mathbf{f}_i$. If $\mathbf{x} \in \mathbb{R}^n$, and $\mathbf{x} \cdot \mathbf{f}_i = 0$ for each i, then show $\mathbf{x} = \mathbf{0}$.

Orthogonal sets and the expansion theorem

Definition: Orthogonal and orthonormal sets A set of vectors $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ is called an $\boxed{\text{orthogonal set}}$ if the set does not contain the zero vector, and $\mathbf{x}_i \cdot \mathbf{x}_j = 0$ for $i \neq j$. A set $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ is called $\boxed{\text{orthonormal}}$ if it is orthogonal and if in addition, $\|\mathbf{x}_i\| = 1$ for each i.

Note that $\{\mathbf{x}\}$ is an orthogonal set if $\mathbf{x} \neq \mathbf{0}$. The standard basis $\{\mathbf{e}_1, \cdots, \mathbf{e}_n\}$ (see Standard basis) is an important orthonormal set in $\mathbb{R}^n$.

Orthogonal sets and bases

Activity

Consider the following vectors in $\mathbb{R}^4$, written as column vectors:

$$\mathbf{f}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix} \quad \mathbf{f}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 2 \end{bmatrix} \quad \mathbf{f}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad \mathbf{f}_4 = \begin{bmatrix} -1 \\ 3 \\ -1 \\ 1 \end{bmatrix}.$$

Verify that these vectors form an orthogonal set in $\mathbb{R}^4$. Then find the orthonormal set obtained by 'normalizing' these vectors, i.e. multiplying each vector by an appropriate constant to obtain a unit vector.

Theorem: Pythagoras' theorem If $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_k\}$ is an orthogonal set in $\mathbb{R}^n$, then

$$\|\mathbf{f}_1 + \mathbf{f}_2 + \cdots + \mathbf{f}_k\|^2 = \|\mathbf{f}_1\|^2 + \|\mathbf{f}_2\|^2 + \cdots + \|\mathbf{f}_k\|^2.$$

Theorem Every orthogonal set in $\mathbb{R}^n$ is linearly independent.

Proof Let $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_k\}$ be an orthogonal set in $\mathbb{R}^n$. To show the set is linearly independent (recalling Linear Independence), we must show the only solution to the linear equation

$$t_1\mathbf{f}_1 + t_2\mathbf{f}_2 + \cdots + t_k\mathbf{f}_k = \mathbf{0}$$

occurs when $t_1 = \cdots = t_k = 0$.
If we take the dot product of both sides of the equation with $\mathbf{f}_1$, we find that

$$\mathbf{f}_1 \cdot (t_1\mathbf{f}_1 + t_2\mathbf{f}_2 + \cdots + t_k\mathbf{f}_k) = \mathbf{f}_1 \cdot \mathbf{0}$$

and therefore that

$$t_1(\mathbf{f}_1 \cdot \mathbf{f}_1) + t_2(\mathbf{f}_1 \cdot \mathbf{f}_2) + \cdots + t_k(\mathbf{f}_1 \cdot \mathbf{f}_k) = 0$$

Since the set of vectors is orthogonal, $\mathbf{f}_1 \cdot \mathbf{f}_j = 0$ for all $j \neq 1$, and $\mathbf{f}_1 \cdot \mathbf{f}_1 = \|\mathbf{f}_1\|^2$, so we conclude that

$$t_1\|\mathbf{f}_1\|^2 = 0.$$

Since $\mathbf{f}_1 \neq \mathbf{0}$ (by definition, an orthogonal set cannot contain the zero vector), $\|\mathbf{f}_1\|^2 > 0$, and so it must be true that $t_1 = 0$. But now repeating this argument with 1 replaced by any index i, we conclude that $t_i = 0$. Therefore the set $\{\mathbf{f}_1, \dots, \mathbf{f}_k\}$ is linearly independent.

Theorem: Expansion theorem Let $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_m\}$ be an orthogonal basis of a subspace U of $\mathbb{R}^n$. If $\mathbf{v}$ is any vector in U, we have

$$\mathbf{v} = \left(\frac{\mathbf{v} \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2}\right)\mathbf{f}_1 + \left(\frac{\mathbf{v} \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2}\right)\mathbf{f}_2 + \cdots + \left(\frac{\mathbf{v} \cdot \mathbf{f}_m}{\|\mathbf{f}_m\|^2}\right)\mathbf{f}_m$$

Proof Since $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_m\}$ spans U, any $\mathbf{v} \in U$ can be written uniquely as

$$\mathbf{v} = t_1\mathbf{f}_1 + t_2\mathbf{f}_2 + \cdots + t_m\mathbf{f}_m$$

for some $t_1, \dots, t_m$. To find coefficient t_1, take the dot product of both sides with $\mathbf{f}_1$, obtaining that

$$\begin{aligned} \mathbf{v} \cdot \mathbf{f}_1 &= (t_1\mathbf{f}_1 + t_2\mathbf{f}_2 + \cdots + t_m\mathbf{f}_m) \cdot \mathbf{f}_1 \\ &= t_1(\mathbf{f}_1 \cdot \mathbf{f}_1) + t_2(\mathbf{f}_2 \cdot \mathbf{f}_1) + \cdots + t_m(\mathbf{f}_m \cdot \mathbf{f}_1). \end{aligned}$$

Since the basis is orthogonal, $\mathbf{f}_i \cdot \mathbf{f}_j = 0$ for $i \neq j$, and so the equation simplifies to

$$\mathbf{v} \cdot \mathbf{f}_1 = t_1\|\mathbf{f}_1\|^2.$$

Rearranging, we conclude that

$$t_1 = \frac{\mathbf{v} \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2}$$

A similar argument shows that

$$t_i = \frac{\mathbf{v} \cdot \mathbf{f}_i}{\|\mathbf{f}_i\|^2}$$

for each i, concluding the proof.

Definition: The Fourier expansion in an orthogonal basis Given an orthogonal basis $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$ for some subspace V of $\mathbb{R}^n$, the expansion of $\mathbf{v} \in V$ as a linear combination of this basis is called the

Fourier expansion

of the vector $\mathbf{v}$, and the coefficients

$$t_i = \frac{\mathbf{v} \cdot \mathbf{f}_i}{\|\mathbf{f}_i\|^2}$$

are called the

Fourier coefficients

of $\mathbf{v}$.

Activity

Let $\mathbf{v} = (a, b, c, d)$ be a vector in $\mathbb{R}^4$. Find the Fourier expansion of $\mathbf{v}$ as a linear combination of the orthogonal basis $\{\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3, \mathbf{f}_4\}$ given in Activity act-6-1-orthogonal-sets.

Dot products and orthogonality now become tools for finding closest vectors.